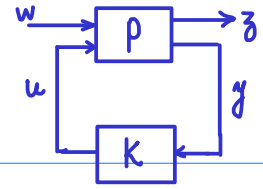
L36

18/11/2020



H_2-control problem
(LQG problem)

Problem: Design a controller K such that

(i) The closed loop system is internally stable.

(ii) The H_2-norm of the transfer matrix $w \longmapsto z$ is minimized.

Why H_2?

① $w \longrightarrow \boxed{G} \longrightarrow z$ $\qquad S_z(j\omega) = |G(j\omega)|^2 S_w(j\omega)$

Suppose w is white noise, $S_w(j\omega) = 1 \Rightarrow S_z(j\omega) = |G(j\omega)|^2$

RMS value of z $\qquad z_{RMS}^2 := \lim_{T\to\infty} \frac{1}{T}\int_0^T z(t)^T z(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_z(j\omega)\,d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} |G(j\omega)|^2 d\omega = \|G\|_{H_2}$

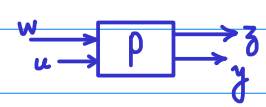
② $\|G\|_{H_2}^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{trace}\left(G^*(j\omega)G(j\omega)\right)d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}\sum_{i=1}^{p} \sigma_i\left(G(j\omega)\right)^2 d\omega$

$G(j\omega) = U(j\omega)\Sigma(j\omega)V^*(j\omega)$

$\text{trace}(G^*G) = \text{trace}(V\Sigma^* U^* U \Sigma V^*)$
$= \text{trace}(V\Sigma^*\Sigma V^*)$
$= \text{trace}(\Sigma^*\Sigma V^* V)$
$= \text{trace}(\Sigma^*\Sigma)$
$= \sum \sigma_i^2$

H_2-norm is concerned with the overall performance over all frequencies

$$\boxed{\begin{aligned} \dot{x} &= Ax + B_1 w + B_2 u \\ z &= C_1 x + D_{11} w + D_{12} u \\ y &= C_2 x + D_{21} w + D_{22} u \end{aligned}}$$

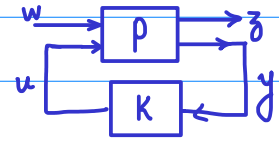


$\Downarrow$

$$\left.\begin{aligned} \dot{x} &= Ax + [B_1 \quad B_2]\begin{bmatrix} w \\ u \end{bmatrix} \\ z &= C_1 x + [D_{11} \quad D_{12}]\begin{bmatrix} w \\ u \end{bmatrix} \\ y &= C_2 x + [D_{21} \quad D_{22}]\begin{bmatrix} w \\ u \end{bmatrix} \end{aligned}\right\} \qquad \begin{aligned} z &= \left\{C_1(sI-A)^{-1}[B_1 \quad B_2] + [D_{11} \quad D_{12}]\right\}\begin{bmatrix} w \\ u \end{bmatrix} \\ y &= \left\{C_2(sI-A)^{-1}[B_1 \quad B_2] + [D_{21} \quad D_{22}]\right\}\begin{bmatrix} w \\ u \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} z \\ y \end{bmatrix} = \begin{bmatrix} \underbrace{C_1(sI-A)^{-1}B_1 + D_{11}}_{P_{11}} & \underbrace{C_1(sI-A)^{-1}B_2 + D_{12}}_{P_{12}} \\ \underbrace{C_2(sI-A)^{-1}B_1 + D_{21}}_{P_{21}} & \underbrace{C_2(sI-A)^{-1}B_2 + D_{22}}_{P_{22}} \end{bmatrix}\begin{bmatrix} w \\ u \end{bmatrix}$$

$$* \quad \overset{w \mapsto z}{=} \quad \mathcal{F}_\ell(P,K) = P_{11} + P_{12}(I - K P_{22})^{-1} K P_{21}$$



Problem: Design a controller K such that

(i) The closed loop system is internally stable.

(ii) $\min_K \| F_\ell(P,K) \|_{\mathcal{H}_2} \longleftarrow F_\ell(P,K) \in R\mathcal{H}_2$ (set of all real, rational, stable, strictly proper t.m)

(a) $P_{11} = C_1(sI - A)^{-1}B_1 + D_{11}$ needs to be strictly proper, i.e. $D_{11} = 0$

(b) D_{22} needs to be zero (not necessary) $\rightarrow$ well-posedness

(c) If K is designed such that K is strictly proper then $\| F_\ell(P,K) \|_{\mathcal{H}_2}$ is well-defined.

Theorem: Suppose G is a system with a state-space realization

$$G := \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array}\right]$$

for the T.M to be in $\mathcal{H}_2$

to ensure well-posedness (this is a sufficiency cond^n)

and (1) (A, B_1) is stabilizable and (A, C_1) is detectable

(2) (A, B_2) is stabilizable and (A, C_2) is detectable

For simplication $\left\{ \begin{array}{l} * \quad (3) \quad D_{12}^T [C_1 \quad D_{12}] = [0 \quad I] \\ * \quad (4) \quad D_{21} [B_1^T \quad D_{21}^T] = [0 \quad I] \end{array} \right.$

Then the optimal stabilizing controller for the $\mathcal{H}_2$-synthesis problem is

$$K = \left[\begin{array}{c|c} A + B_2 F + L C_2 & -L \\ \hline F & 0 \end{array}\right]$$

$$\dot{x}_K = A_K x_K + B_K y$$
$$u = C_K x_K + D_K y$$

with $L := -Y C_2^T$ and $F := -B_2^T X$ where $X = \mathrm{Ric}(J_1)$ and $Y = \mathrm{Ric}(J_2)$,

where $\underline{\underline{J_1}} := \begin{bmatrix} A & -B_2 B_2^T \\ -C_1^T C_1 & -A^T \end{bmatrix}$ and $\underline{\underline{J_2}} := \begin{bmatrix} A^T & -C_2^T C_2 \\ -B_1 B_1^T & -A \end{bmatrix}$

Algebraic Riccati Equation (ARE):

$$\boxed{ax^2 + bx + c = 0}$$

$$\boxed{A^T X + XA + XMX + Q = 0}$$

where $Q = Q^T \in \mathbb{R}^{n\times n}$, $M = M^T \in \mathbb{R}^{n\times n}$, X variable.
$A \in \mathbb{R}^{n\times n}$

Let us define a matrix H corresponding to the ARE

$$H := \begin{bmatrix} A & M \\ -Q & -A^T \end{bmatrix} \in \mathbb{R}^{2n\times 2n} \longleftarrow \text{Hamiltonian matrix}$$

Properties of H:- ① If $\lambda \in \text{spec}(H)$, then $-\lambda \in \text{spec}(H)$

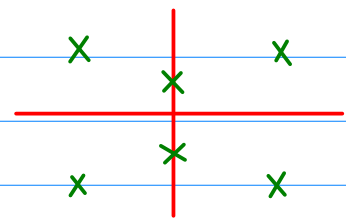
$$J := \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}$$

$$J^{-1} H J = -H^T$$

λ_1 $\quad$ λ_1

$-\lambda_1 \leftarrow -\lambda_1 \in \text{spec}(-H^T)$

symmetric eigenvalues about real & imag. axis



$$\#\,\text{spec}(H) = 2n \to \mathbb{C}^+,\ j\mathbb{R},\ \mathbb{C}^-$$

Solution of ARE: Let us define two matrices $X_1 \in \mathbb{R}^{n\times n}$, $X_2 \in \mathbb{R}^{n\times n}$ such that $\text{img}\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is a n-dimensional invariant subspace of H.

$$H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \Lambda, \quad \Lambda \in \mathbb{R}^{n\times n}$$

$$(\ ) = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \alpha$$

$$\boxed{H\mathcal{V} \subseteq \mathcal{V}}$$

If we assume X_1 to be invertible, then

$$H \begin{bmatrix} I \\ X_2 X_1^{-1} \end{bmatrix} X_1 = \begin{bmatrix} I \\ X_2 X_1^{-1} \end{bmatrix} X_1 \Lambda$$

Define $X_2 X_1^{-1} := X$

If $\text{img}\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is H-inv then $\text{img}\begin{bmatrix} I \\ X \end{bmatrix}$ is also H invariant

$$\begin{bmatrix} A & M \\ -Q & -A^T \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} = \begin{bmatrix} I \\ X \end{bmatrix} X_1 \Lambda X^{-1} \longrightarrow ①$$

Multiply $[-X \quad I]$ from left in eqn ①

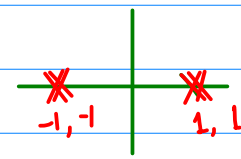
$$\begin{bmatrix} -X & I \end{bmatrix} \begin{bmatrix} A & M \\ -Q & -A^T \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix} = \underbrace{\begin{bmatrix} -X & I \end{bmatrix} \begin{bmatrix} I \\ X \end{bmatrix}}_{0} X_1 \Lambda X_1^{-1}$$

$$\Rightarrow \quad -XA - Q - XMX - A^TX = 0 \quad \Rightarrow \quad \boxed{A^TX + XA + XMX + Q = 0}$$

X is a solution of the ARE.

$n=2$

e.g. $A = \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix}$, $M = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$



$$\begin{array}{c} n \\ 2n \end{array} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \quad \boxed{4 \times 2}$$

$$\begin{array}{cccc} -1 & -1 & 1 & 1 \\ \end{array}$$

$$\underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3/2 \\ 0 \\ 0 \end{bmatrix}}, \quad \underbrace{\begin{bmatrix} 1 \\ 2 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ -3/2 \\ 1 \\ 0 \end{bmatrix}}$$

$$H \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \underline{\underline{\Lambda}}$$

$$H v = \lambda v$$

$$H \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \lambda_1 v_1 \quad \lambda_2 v_2$$

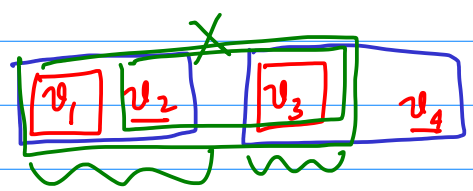
$$= \underbrace{\begin{bmatrix} v_1 & v_2 \end{bmatrix}} \underline{\underline{\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}}}$$

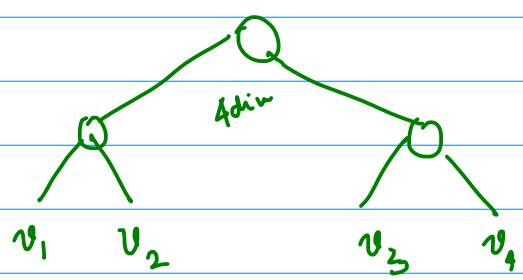
$$X_1 = \begin{bmatrix} 1 & -1 \\ 2 & -3/2 \end{bmatrix} \qquad X_2 = \begin{bmatrix} 2 & 1 \\ -2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 3/2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{matrix} \left.\begin{matrix} \\ \\ \end{matrix}\right\} \Rightarrow X_1 \\ \left.\begin{matrix} \\ \\ \end{matrix}\right\} \Rightarrow X_2 \end{matrix} \qquad X = X_2 X_1^{-1} = 0$$

$$\underline{A^TX} + \underline{XA} + \underline{XMX} + \underline{Q} = 0$$

H





$\boxed{v_1 \quad v_3} \quad v_2$

$$\boxed{A \begin{bmatrix} v_1 & v_3 \end{bmatrix} = \lambda v_1 \quad \lambda v_3 + v_1}$$